Difference method for generation of circular arcs and ellipses

W P Wang and C Y Wang

A method, called the difference method, is presented to generate the vertices of the inscribed polygon of a circular arc or an elliptic arc. Only two multiplications are needed by the algorithm to compute one vertex of the polygon. Elaborate analysis of the error behaviour and experimental results are provided to show that the difference method is stable. Finally, an example of its application and a possible extension of the method are discussed.

geometry, shape generation, ellipses, difference method

Circle drawing is an important topic in computer graphics. Many techniques have been developed for this problem[1,2]. They are usually classified into two kinds. The first kind focuses on how to approximate the arc with grid points nearest to the arc on the plane, of which Bresenham's algorithm for circle drawing is typical. The second kind of method is to approximate the arc with its inscribed polygon, which is subsequently drawn as a sequence of straight line segments, of which the matrix method, or incremental rotation method, is a representative. The latter approach will be concentrated on in the following discussion.

For drawing a circular arc, the well known matrix method and some of its variants have been widely employed to generate the inscribed polygon of the arc[3]. It generally requires four multiplications for the matrix method to compute a new vertex of the polygon. In this paper, a new method in the form of the difference equation is introduced to generate the inscribed polygon of a circular arc. The method is extended to the generation of ellipses.

The main results of this paper include:

- difference method which needs only two multiplications to compute a new vertex of the inscribed polygon
- analysis of the stability of the method

Finally, numerical examples, an example of the application and suggestions for further extension of the difference method are provided.

GENERATION OF CIRCULAR ARCS

Throughout this paper, vector $\mathbf{P}=(x, y)^{\mathrm{T}}$ is used to denote the point $P(x, y)$ in E^2. This section starts by briefly reviewing the matrix method to provide a comparison with the new method. Given a point $\mathbf{P}_0=(x_0, y_0)^{\mathrm{T}}$, a point sequence $\{\mathbf{P}_i\}_{i\geqslant 0}$ is derivable from the following recurrence formula

$$\mathbf{P}_{i+1}=\mathbf{T}\mathbf{P}_i \qquad i\geqslant 0 \tag{1}$$

where $\mathbf{T}$ is a rotation matrix,

$$\mathbf{T}=\begin{pmatrix}\cos\beta & -\sin\beta\\ \sin\beta & \cos\beta\end{pmatrix}$$

It is easy to show that $\{\mathbf{P}_i\}_{i\geqslant 0}$ is an equidistance point sequence on a circle centred at the origin with radius $R=|\mathbf{P}_0|$, and four multiplications are needed for equation (1) to compute a new point. When a homogeneous affine transformation exists, a circle is generally mapped onto an ellipse. Using $\{\mathbf{Q}_i\}_{i\geqslant 0}$ to denote the point sequence on the ellipse, then equation (1) will be changed as

$$\mathbf{Q}_{i+1}=\mathrm{M}\mathbf{T}\mathrm{M}^{-1}\mathbf{Q}_i \qquad i\geqslant 0 \tag{2}$$

where $\mathbf{Q}_i=\mathrm{M}\mathbf{P}_i$. This means that the matrix involved in the computation has to be changed if an affine transformation exists, which is inconvenient for practical application. Cohen has extended this approach to the display of general conics[4].

Difference method

Theorem 1

Suppose $\mathbf{P}_0$ and $\mathbf{P}_1$ are two distinct points on circle C which is centred at the origin, β is the centre angle subtended by arc $\mathbf{P}_0\mathbf{P}_1$ and $\{\mathbf{P}_i\}_{i\geqslant 0}$ are generated by the following recurrence formula

$$\mathbf{P}_{i+2}=2\cos\beta\mathbf{P}_{i+1}-\mathbf{P}_i \qquad i\geqslant 0 \tag{3}$$

then $\{\mathbf{P}_i\}_{i\geqslant 0}$ are the successive vertices of the inscribed polygon of circle C, and $\angle\,\mathbf{P}_i\mathrm{O}\mathbf{P}_{i+1}=\beta$, $i\geqslant 0$.

Proof

Suppose circle C is defined by the following trigonometric parametric equation

$$\begin{cases}x=R\sin t\\ y=R\cos t\end{cases} \qquad t\in(-\infty, \infty)$$

Department of Computer Science, Shandong University, Jinan, Shandong, People's Republic of China

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where R is the radius.

Let t_0, $t_0 + \beta$ be the values of parameter t for $\mathbf{P}_0$ and $\mathbf{P}_1$ respectively, then

$$\mathbf{P}_0 = [R \cos t_0, R \sin t_0]^T$$

$$\mathbf{P}_1 = [R \cos (t_0 + \beta), R \sin (t_0 + \beta)]^T$$

Suppose for $\mathbf{P}_i$ produced by equation (3), $\mathbf{P}_i = [R \cos (t_0 + i\beta), R \sin (t_0 + i\beta)]^T$ for all $i \leqslant m + 1$. Appealing to the following trigonometric identities

$$\cos (t + 2\beta) = 2 \cos \beta \cos (t + \beta) - \cos t$$

$$\sin (t + 2\beta) = 2 \cos \beta \sin (t + \beta) - \sin t$$

obtains

$$\begin{aligned}\mathbf{P}_{m+2} &= 2 \cos \beta \mathbf{P}_{m+1} - \mathbf{P}_m \\ &= \begin{bmatrix} 2 \cos \beta R \cos[t_0 + (m+1)\beta] - R \cos (t_0 + m\beta) \\ 2 \cos \beta R \sin [t_0 + (m+1)\beta] - R \sin (t_0 + m\beta) \end{bmatrix} \\ &= [R \cos \beta[t_0 + (m+2)\beta], R \sin [t_0 + (m+2)\beta]^T\end{aligned}$$

By induction, it has been proved that for all $i \geqslant 0$

$$\mathbf{P}_i = [R \cos (t_0 + i\beta), R \sin (t_0 + i\beta)]^T \tag{4}$$

This completes the proof of Theorem 1 QED.

The fact that equation (4) satisfies equation (3) was also mentioned in Henrie[5]. Figure 1 provides an illustration of Theorem 1 from a geometric point of view. According to Theorem 1, given two initial points on the intended circle, equation (3) can be used to generate the equidistance sequence $\{\mathbf{P}_i\}_{i \geqslant 0}$ on the circle. Compared with the matrix method that requires four multiplications for each step, the difference method needs only two multiplications, so it is more efficient. Note that the difference method needs two initial points to start the computation; so if only the starting point is given, the matrix method should be applied to obtain the second point, then the difference method can be used.

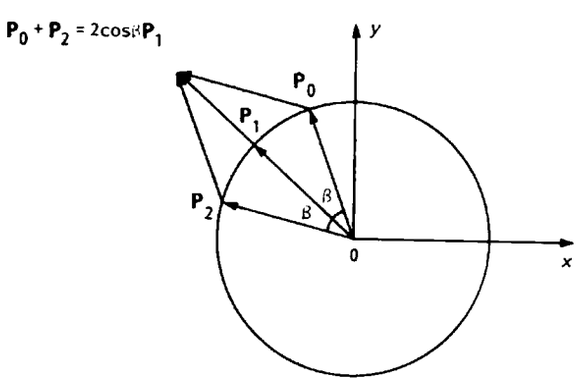

Figure 1. Vector $\mathbf{P}_1$ is collinear with $\mathbf{P}_0 + \mathbf{P}_2$

GENERATION OF ELLIPSES

Theorem 2

Given an ellipse E centred at the origin, a starting point $\mathbf{Q}_0$ on E and a real constant k, $|k| < 1$, there exists point $\mathbf{Q}_1$ on E such that the point sequence $\{\mathbf{Q}_i\}_{i \geqslant 0}$ generated by

$$\mathbf{Q}_{i+2} = 2k\mathbf{Q}_{i+1} - \mathbf{Q}_i \qquad i \geqslant 0 \tag{5}$$

lie on E.

Proof

It is known that E can be mapped from the unit circle C by some homogeneous affine transformation M (Ahuja and Coons[6]). Use $\mathbf{M}$ to denote the matrix of M with respect to some orthogonal basis, choose β such that $\cos \beta = k$, then let $\mathbf{P}_0 = \mathrm{M}^{-1}\mathbf{Q}_0$, $\mathbf{P}_1 = \mathbf{T}\mathbf{P}_0$, where

$$\mathbf{T} = \begin{bmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{bmatrix}$$

Then by Theorem 1, it follows that the point sequence generated by

$$\mathbf{P}_{i+2} = 2 \cos \beta \mathbf{P}_{i+1} - \mathbf{P}_i \qquad i \geqslant 0 \tag{6}$$

lie on the unit circle C. Multiply $\mathbf{M}$ by both sides of equation (6), and denote $\mathbf{Q}_i = \mathbf{M}\mathbf{P}_i$, $i \geqslant 0$, where obviously $\mathbf{Q}_i$ lie on E, gives

$$\mathbf{Q}_{i+2} = 2 \cos \beta \mathbf{Q}_{i+1} - \mathbf{Q}_i \qquad i \geqslant 0 \tag{7}$$

It is now clear that if $\mathbf{Q}_1 = \mathbf{M}\mathbf{P}_1$, then the point sequence generated by equation (7) belongs to E. This completes the proof of Theorem 2 QED.

Theorem 2 indicates that equation (3) keeps its form invariant under homogeneous affine transformation. This desirable property enables an ellipse to be drawn by using the same formula (3) as a circle without the inverse matrix that appears in (2). Here again, two multiplications suffice to compute a new point on the ellipse.

Since the difference method is intrinsically a two-step method, the choice of the initial points $\mathbf{Q}_0$, $\mathbf{Q}_1$ on the ellipse should be consistent with that of parameter k. Once the first point is given, the proof of Theorem 2 also provides the way to yield the second point. However, it is worth noting that $\beta = \pm \arccos k$; the choice of the sign determines the direction in which the generated point sequence moves on the ellipse.

So far, it has been shown that equation (3) can serve to generate the inscribed polygon of both circular arcs and ellipses with only two multiplications for each step. In practice, when the algorithm is implemented in assembly codes in a microprocessor, particularly where $k = 1 - 2^{-N}$ (N is some positive integer), the speed-up of the computation will be more remarkable, since the calculation now needs nothing more than simple additions and shifts.

ERROR ANALYSIS

In the following discussion, the propagation of initial errors only is studied first. The case that a rounding error is introduced in the computing of each vertex is considered by using the derived result of the former case. The initial errors of $\mathbf{P}_0, \mathbf{P}_1$ are denoted by $\bar{f}_0$ and $\bar{f}_1$ respectively, and the error of $\mathbf{P}_i$ is $\bar{f}_i$. Let $\bar{\mathbf{P}}_i$ be the point with the error $\bar{f}_i$, i.e. $\bar{\mathbf{P}}_i = \mathbf{P}_i + \bar{f}_i$, then in actual calculation, $\{\bar{\mathbf{P}}_n\}_{n \geqslant 0}$ are generated by

$$\bar{\mathbf{P}}_{n+2} = 2\cos\beta\bar{\mathbf{P}}_{n+1} - \bar{\mathbf{P}}_n \qquad n \geqslant 0 \tag{8}$$

On the other hand

$$\mathbf{P}_{n+2} = 2\cos\beta\mathbf{P}_{n+1} - \mathbf{P}_n \qquad n \geqslant 0 \tag{9}$$

Subtracting equation (9) from equation (8), noting that $\bar{f}_n = \bar{\mathbf{P}}_n - \mathbf{P}_n$, gives

$$\bar{f}_{n+2} = 2\cos\beta\bar{f}_{n+1} - \bar{f}_n \qquad n \geqslant 0 \tag{10}$$

Theorem 3
The solutions to equation (10) are

$$\bar{f}_n = \{\sin(n\beta)\bar{f}_1 - \sin[(n-1)\beta]\bar{f}_0\}/\sin\beta$$

Proof
The eigenequation of (10) is

$$v^2 - 2\cos\beta v + 1 = 0$$

whose roots are $v_{1,2} = \cos\beta \pm i\sin\beta = e^{\pm i\beta}$. According to the theory of difference equation, the solutions of equation (10) are

$$\bar{f}_n = [(v_1^n - v_2^n)\bar{f}_1 - (v_1^{n-1} - v_2^{n-1})\bar{f}_0/(v_1 - v_2) \qquad n \geqslant 0 \tag{11}$$

Substituting $v_1 = e^{i\beta}$, $v_2 = e^{-i\beta}$ in equation (11), gives

$$\begin{aligned}\bar{f}_n &= [(e^{in\beta} - e^{-in\beta})\bar{f}_1 \\ &\quad - (e^{i(n-1)\beta} - e^{-i(n-1)\beta})\bar{f}_0]/(e^{i\beta} - e^{-i\beta}) \\ &= [(e^{in\beta} - e^{-in\beta})/(2i)\bar{f}_1 \\ &\quad - (e^{i(n-1)\beta} - e^{-i(n-1)\beta})/(2i)\bar{f}_0]/[(e^{i\beta} - e^{-i\beta})/(2i)] \\ &= \{\sin(n\beta)\bar{f}_1 - \sin[(n-1)\beta]\bar{f}_0\}/\sin\beta \qquad \text{QED}\end{aligned}$$

From Theorem 3 it can be learnt that the error $\bar{f}_n$, which is caused by $\bar{f}_1$ and $\bar{f}_0$, is bounded by $(\|\bar{f}_1\| + \|\bar{f}_0\|)/\sin\beta$, and $n(\|\bar{f}_1\| + \|\bar{f}_0\|)$ is another bound to be used in later analysis.

The rounding error occurring at each step will now be taken into account, and the initial errors will be ignored. Suppose that r_n is the error introduced when $\mathbf{P}_n$ is computed by equation (3) and f_n denotes the error of $\mathbf{P}_n$, then the error due to the imprecision of $\cos\beta$ could also be contained in r_n. Thus giving

$$\bar{\mathbf{P}}_0 = \mathbf{P}_0$$

$$\bar{\mathbf{P}}_1 = \mathbf{P}_1$$

$$\bar{\mathbf{P}}_{n+2} = 2\cos\beta\bar{\mathbf{P}}_{n+1} - \bar{\mathbf{P}}_n + r_{n+2} \qquad n \geqslant 0 \tag{12}$$

Subtracting equation (9) from (12), since $f_n = \bar{\mathbf{P}}_n - \mathbf{P}_n$ gives

$$f_{n+2} = 2\cos\beta f_{n+1} - f_n + r_{n+2} \qquad n \geqslant 0 \tag{13}$$

Theorem 4
The error f_n determined by equation (13) is

$$f_n = \left\{\sum_{m=2}^{n} \sin[(n-m+1)\beta]r_{n-m}\right\} \Big/ \sin\beta \qquad n \geqslant 0$$

Proof
Equation (13) is divided into $n-1$ separate systems of difference equations (which are easier to solve) to find out f_n. It is easy to show that the sum of the solutions to all these $n-1$ systems gives the solution to Equation (13). Define

$$r_{j+2}^{(m)} = \begin{cases} r_{j+2} & \text{when } m = j+2 \\ 0 & \text{otherwise} \end{cases}$$

Then the $n-1$ systems are as follows

$$\begin{cases} f_0^{(m)} = 0 \\ f_1^{(m)} = 0 \\ f_{i+2}^{(m)} = 2\cos\beta f_{i+1}^{(m)} - f_i^{(m)} + r_{i+2}^{(m)} & i = 0, 1, 2, \ldots, n-2 \\ & m = 2, 3, \ldots, n \end{cases}$$

which are equivalent to

$$\begin{cases} f_m^{(m)} = r_m \\ f_{m+1}^{(m)} = 2\cos\beta r_m \\ f_{j+2}^{(m)} = 2\cos\beta f_{j+1}^{(m)} - f_j^{(m)} & j = m, m+1, \ldots, n-2 \\ & m = 2, 3, \ldots, n \end{cases}$$

According to Theorem 3, the solutions to all these $n-1$ systems are

$$f_n^{(m)} = \sin[(n-m+1)\beta]r_m/\sin\beta \qquad m = 2, 3, \ldots, n-2$$

The solution to equation (3) is therefore

$$f_n = \sum_{m=2}^{n} f_n^{(m)} = \sum_{m=2}^{n} \sin[(n-m+1)\beta]r_m/\sin\beta \qquad \text{QED}$$

Let $D = \max_{2 \leqslant m \leqslant n}\{\|r_m\|\}$, and assume β is a small angle, then $\sin\beta \approx \beta$. By Theorem 4, noting that $|\sin[(n-m+1)\beta]| \leqslant 1$, obtain

$$\|f_n\| \leqslant nD/\beta \tag{14}$$

If the initial errors are also considered, the estimate of the total error h_n is

$$\|h_n\| \leqslant \|f_n\| + \|\bar{f}_n\| \leqslant nD/\beta + n(\|\bar{f}_1\| + \|\bar{f}_0\|) \tag{15}$$

The dependence of the total error h_n on the radius R of the circle is investigated. When a circular arc is generated, it is usually required that the tolerance

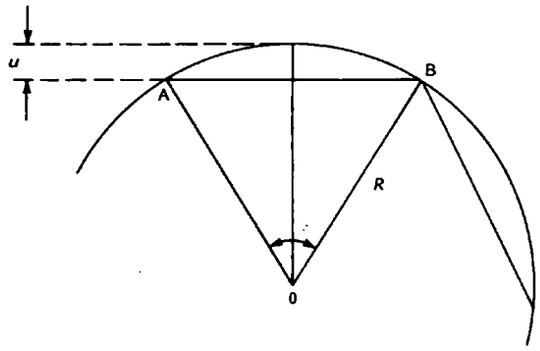

Figure 2. AB is one side of the inscribed polygon of circle 0, R is the radius, and β is the centre angle subtended by arc AB

between the approximating chord and the arc subtended does not exceed a constant u (see Figure 2). This means $R - R\cos(\beta/2) \approx u$, consequently, $\beta \approx (8u/R)^{1/2}$, from which and from formula (15), it follows that

$$\|h_n\| \leqslant nD(R/8u)^{1/2} + n(\|\bar{f}_1\| + \|\bar{f}_0\|)$$

Let d be the relative precision of the computer in use, and assume that $\|\bar{\mathbf{P}}_i\| = O(R)$, then $O(R)d$ is the reasonable estimate of D, $\|\bar{f}_0\|$ and $\|\bar{f}_1\|$. Let L be the length of the arc intended to draw, then

$$n = L/(R\beta) = L/(8uR)^{1/2}$$

therefore

$$\|h_n\| = (L/(8uR)^{1/2})O(R)d((R/8u)^{1/2}) + 2(L/(8uR)^{1/2})O(R)d = O(dLR/u) \qquad (16)$$

Experimental results corresponding to above analysis are listed in Tables 1 and 2. In Table 1, entire circles are drawn and the maximal errors are listed; in Table 2, arcs of fixed length L but with various radii are considered. In these examples the tolerance u is assumed to be 0.5. The algorithm was programmed in C language, and executed under Unix on a VAX/750.

The analysis summarized in equation (16) is well confirmed by these two groups of examples. In Table 1, it is just necessary to note the fact that the arc length L is proportional to the radius R.

Table 1. Entire circles $x^2 + y^2 = R^2$ are drawn using difference method. E is the maximal error; N is the number of points computed on the circle

R	N	E
10	10	0.000009
20	15	0.000025
100	32	0.000251
200	45	0.001429
1 000	100	0.033671
2 000	141	0.135532
10 000	315	3.177154
20 000	445	11.599296

Table 2. Arcs of the circles $x^2 + y^2 = R^2$ with fixed length $L = 10\,000$ are drawn

R	L	N	E
1 000	10 000	159	0.056414
2 000	10 000	112	0.124219
5 000	10 000	71	0.463366
10 000	10 000	51	0.477770
20 000	10 000	36	0.445330
50 000	10 000	23	0.121287
100 000	10 000	16	0.922545
200 000	10 000	12	0.125381
500 000	10 000	8	0.002930
1 000 000	10 000	5	1.312518

APPLICATION

Besides its use in general graphic packages for arc generation where the matrix method has conventionally been adopted, the difference method can also apply to many other situations where speed is emphasized. Its application in a microprocessor-driven high-speed intelligent plotter is illustrated next.

The difference method has been successfully implemented in the design of a high-speed intelligent plotter[7], in which two microprocessors (Motorola MPU 6800) are matched together to work in pipeline mode as the plotter controller. To generate a circular arc, the vertices of its inscribed polygon are produced by MPU_1 and then transmitted to MPU_2, where the segments between adjacent vertices are drawn by the well-known Bresenhem's straight-line drawing algorithm (see Figure 3).

The experimental results lead us to the conclusion that, if the difference method is adopted in MPU_1, the vertices of the polygon can be produced fast enough to match the straight-line drawing speed of the Bresenham's algorithm executed in MPU_2, i.e. the arc can be drawn as fast as a straight-line segment is drawn by the Bresenham's algorithm, but the matrix method fails to achieve this performance, because it requires four multiplications. What is more, when implementing the algorithm in microprocessor, let the constant k be a number of the form $1 - 2^{-N}$ (N is some positive integer) to speed up the computing further, for this means that the computation can be carried out easily by additions and shifts only.

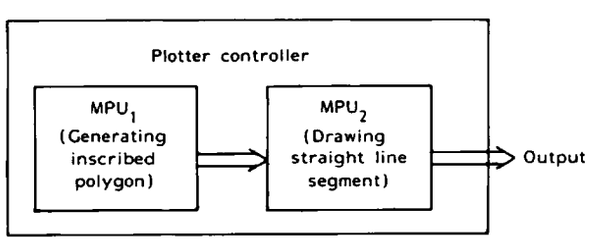

Figure 3. Pipeline mode architecture for generating circular arcs

CONCLUSIONS

Arc drawing is an important subject in computer graphics. Here, a new method has been presented, called the difference method, to generate a sequence of evenly distributed points on a circular arc. This method needs only two multiplications to generate a point, and is therefore more efficient than the traditional matrix method. It has been verified by numerical examples that the former is twice as fast as the latter on average. When the algorithm is programmed in assembly codes, the efficiency of the difference method becomes more notable, as it can be used such that only additions and shifts are required in its computation. It is worth noting that the method naturally extends to the case of elliptic arc generation without change of form or reduction of efficiency.

The careful analysis of the propagated errors and example of the application of the method discussed above show that the difference method is both stable and practicable.

It is well known that difference equation

$$\mathbf{P}_{i+3} = 3\mathbf{P}_{i+2} - 3\mathbf{P}_{i+1} + \mathbf{P}_i \qquad i \geqslant 0 \qquad (17)$$

can be used to generate parabolas. It has also been shown by the authors[8] that hyperbolas may be produced by the following difference equation

$$\mathbf{P}_{i+2} = 2k\mathbf{P}_{i+1} - \mathbf{P}_i \qquad k > 1 \qquad i \geqslant 0 \qquad (18)$$

From equations (3), (17) and (18), it may be concluded that all conic sections can be generated by difference equations. Further, the authors would like to point out that the difference method produces the same polygon as that produced by matrix method, so the result is comparable to that obtained by Cohen[4], and the approximating property of the matrix method proposed by Smith[9] also holds for the difference method. All this will be discussed in another paper[8].

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